

OLG with Production

Diamond Framework

Contents

1. From Exchange to Production
2. Households
3. Production Sector and Factor Markets
4. Equilibrium Dynamics and Stability
5. Cobb-Douglas + Log: Closed-Form Solution
6. The Command Optimum
7. Summary

From Exchange to Production

What Changes When We Add Production?

Lecture 1 recap

Each gen t solves $\max U(c_t^1) + \beta U(c_{t+1}^2)$ s.t. $c_t^1 + s_t = w_1$, $c_{t+1}^2 = (1 + r_{t+1})s_t$, with **fixed** w_1, w_2 and r_{t+1} from $s_t = m/p_t$.

Three things change with production:

1. **Endowments endogenous:** $w_t = \omega(k_t) \equiv f(k_t) - k_t f'(k_t)$ (competitive wage).
2. **Interest rate from capital:** $r_{t+1} = \rho(k_{t+1}) \equiv f'(k_{t+1}) - \delta_K$, not from money. Savings $\rightarrow$ physical capital k_{t+1} .
3. **Capital market clears:** $(1 + n)k_{t+1} = s_t$. Transition map $k_{t+1} = g(k_t)$ replaces the offer-curve cobweb algorithm.

What stays the same

The Euler equation $U'(c_t^1) = \beta(1 + r_{t+1})U'(c_{t+1}^2)$ is identical. The savings function $s(w, r)$ is Lecture 1's $f(r)$ with endogenous (w, r) . All FWT pathologies of Lecture 1 persist — in a new form.

Lecture 1 – 2: Connections

Concept	L1 (exchange)	L2 (production)
Young's endowment	w_1 (exogenous)	$w_t = \omega(k_t)$ (endogenous)
Return on savings	r (money market)	$r_{t+1} = f'(k_{t+1}) - \delta_K$
Savings vehicle	Fiat money / bonds	Physical capital k_{t+1}
Market clearing	$s_t = m/p_t$	$(1+n)k_{t+1} = s_t$
Equilibrium dynamics	Cobweb algorithm	Transition map $g(k_t)$
Samuelson condition	$\bar{r} < 0$	$r^* < n$ (dyn. inefficiency)
Social transfer	Outside money	Capital / PAYG
FWT failure	$\sum_t p_t \bar{e}_t = +\infty$	$k^* > k^{\text{GR}}$
Pareto improvement	PAYG / public debt / money	PAYG / public debt / money
Golden Rule	$r^* = 0$: budget $\equiv$ feasibility	$r^* = n$: budget slope $= -(1+n)$

Every row is a strict generalisation. Setting $f(k) = w_1 + (1+r)k$ (linear) reduces the production model to L1 exactly.

Households

Household Problem (Diamond 1965)

Preferences: $U_t = u(c_t^1) + \beta u(c_{t+1}^2)$, $\beta > 0$, $u' > 0$, $u'' < 0$, Inada.

Budget constraints: $c_t^1 + s_t = w_t$ (young), $c_{t+1}^2 = (1 + r_{t+1}) s_t$ (old).

Euler equation (FOC):

$$\boxed{u'(c_t^1) = \beta (1 + r_{t+1}) u'(c_{t+1}^2)} \Rightarrow s_t = s(w_t, r_{t+1}).$$

Existence and Uniqueness of Optimal Savings

Proposition

For any $w_t > 0$, $r_{t+1} > -1$: $\exists!$ $s_t \in (0, w_t)$ satisfying the Euler equation. Write $s_t = s(w_t, r_{t+1})$.

Proof.

Define $\Phi(s) \equiv u'(w_t - s) - \beta(1 + r_{t+1})u'((1 + r_{t+1})s)$.

Existence. $\Phi(0) = u'(w_t) - \beta(1 + r_{t+1}) \cdot u'(0^+) = -\infty < 0$ (Inada: $u'(0) = +\infty$).
 $\Phi(w_t) = u'(0^+) - \beta(1 + r_{t+1})u'((1 + r_{t+1})w_t) = +\infty > 0$. IVT: $\exists s^* \in (0, w_t)$ with $\Phi(s^*) = 0$.

Uniqueness. $\Phi'(s) = -u''(w_t - s) - \beta(1 + r_{t+1})^2 u''((1 + r_{t+1})s) > 0$ since $u'' < 0$. Φ strictly increasing $\Rightarrow$ at most one zero. □

The Savings Function: Comparative Statics

From implicit differentiation of $\Phi(s) = 0$:

Wage effect: $\frac{\partial s}{\partial w} = \frac{-u''(c^1)}{-u''(c^1) - \beta(1+r)^2 u''(c^2)} \in (0, 1)$. Savings always increase in w .

Interest-rate effect: $\sigma(c) \equiv -u''(c)c/u'(c) > 0$ (inverse IES ζ).

Proposition

$$\frac{\partial s}{\partial r} = \frac{\beta u'(c^2)[1 - \sigma(c^2)]}{-u''(c^1) - \beta(1+r)^2 u''(c^2)}.$$

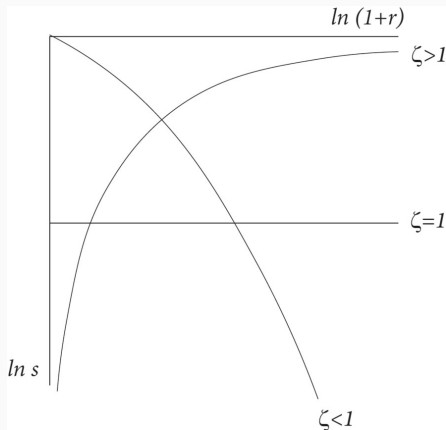
Denominator > 0 , so $\partial s / \partial r \gtrless 0 \iff \sigma(c^2) \lessgtr 1 \iff \text{IES} \gtrless 1$.

Connection to Lecture 1: backward-bending offer curve

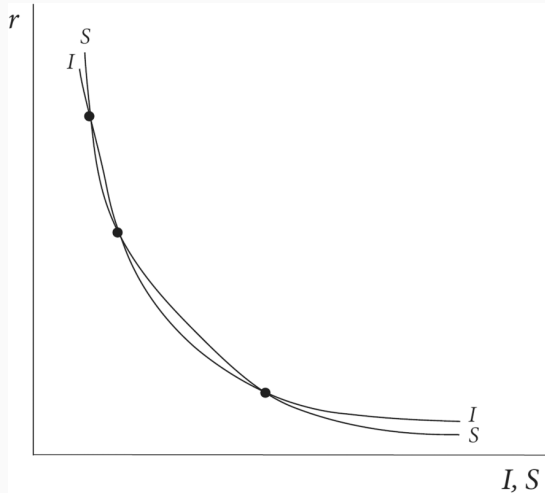
In L1 the offer curve bends backward when $\partial s / \partial r < 0$, i.e. $\zeta < 1$ ($\sigma > 1$) — generating endogenous cycles. Here *exactly the same condition* drives **poverty traps** and non-monotone transition maps in L2.

Log Utility: Closed Form

With $u(c) = \ln c$: $\sigma = \zeta \equiv 1$, $\partial s / \partial r = 0$. From $1/(w-s) = \beta(1+r)/[(1+r)s]$: $s_t = \frac{\beta}{1+\beta} w_t$
— fixed fraction of wage.



Multiplicity in I-S Framework



Production Sector and Factor Markets

CRS production: $Y_t = F(K_t, N_t)$, per worker:

$$y_t = f(k_t), \quad f' > 0, f'' < 0, f'(0) = +\infty, f'(\infty) = 0.$$

Competitive factor prices:

$$w_t = \omega(k_t) \equiv f(k_t) - k_t f'(k_t), \quad r_t = \rho(k_t) \equiv f'(k_t) - \delta_K.$$

Monotonicity

$\omega'(k) = -kf''(k) > 0$: wages $\uparrow$ in k . $\rho'(k) = f''(k) < 0$: returns $\downarrow$ in k . In L1 the endowment w_1 was *fixed* and r was a parameter. Here both are *moving* functions of k_t .

Capital Market Clearing and the Transition Map

Labour: N_t young workers supply inelastically.

Capital: $K_{t+1} = N_t s_t$.

Dividing by $N_{t+1} = (1 + n)N_t$:

$$(1 + n) k_{t+1} = s(\omega(k_t), \rho(k_{t+1})).$$

Fundamental difference equation of Diamond (1965).

Equilibrium Dynamics and Stability

The Transition Map $k_{t+1} = g(k_t)$

By the IFT, $(1+n)k_{t+1} = s(\omega(k_t), \rho(k_{t+1}))$ implicitly defines g wherever $(1+n) - s_r \rho'(k_{t+1}) \neq 0$. Differentiating:

$$g'(k) = \frac{s_w \omega'(k)}{(1+n) - s_r \rho'(g(k))}.$$

Local stability

k^* locally stable $\Leftrightarrow |g'(k^*)| < 1$. More likely when: (i) $IES \geq 1$ ($s_r \geq 0$); (ii) $|f''|$ small; (iii) n large. **$IES < 1$ ($s_r < 0$) can destabilise** — same condition as backward-bending offer curve (Lecture 1) that generates cycles.

Steady States: Existence, Uniqueness, Multiplicity

Case 1 — Unique stable k^*

g crosses 45° line once from above.

Guaranteed with log utility + CD

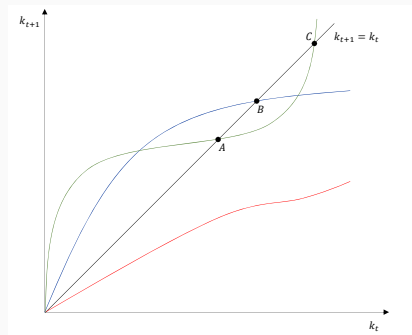
($g' = \alpha < 1$).

Case 2 — Multiple steady states

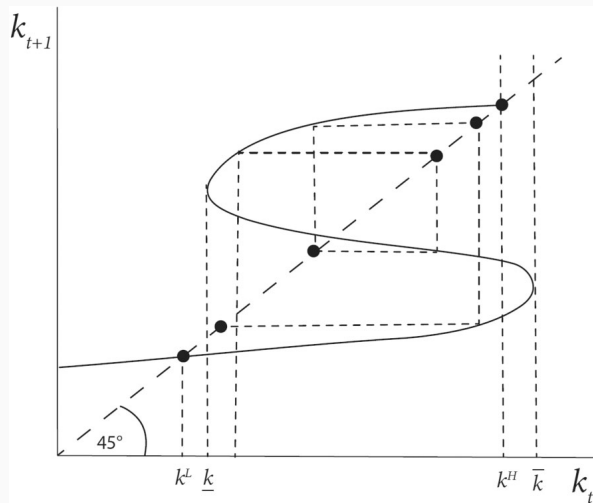
g non-monotone when $s_r < 0$ (IES < 1): two stable $k_L < k_H$, one unstable k_u . **Poverty trap:** start below $k_u \Rightarrow k_L$. *Production-economy counterpart of L1's multiple equilibria.*

Case 3 — No interior SS

$g(k) < k$ or $g(k) > k$ everywhere.



Steady States: Existence, Uniqueness, Multiplicity



Cobb-Douglas + Log: Closed-Form Solution

Parametric Example: Cobb-Douglas + Log Utility

$$f(k) = k^\alpha, \omega(k) = (1 - \alpha)k^\alpha,$$

$$\rho(k) = \alpha k^{\alpha-1} - \delta_K.$$

$$\text{Log: } s_t = \frac{\beta}{1+\beta} \omega(k_t). \text{ Clearing:}$$

$$(1 + n)k_{t+1} = \frac{\beta}{1+\beta} (1 - \alpha)k_t^\alpha.$$

Transition map

$$k_{t+1} = \underbrace{\frac{\beta(1 - \alpha)}{(1 + \beta)(1 + n)}}_A k_t^\alpha \equiv g(k_t)$$

$$k^* = A^{1/(1-\alpha)}. \quad g'(k^*) = \alpha < 1. \quad \checkmark$$

Calibration ($\delta_K = 0$, period = 30 yr):

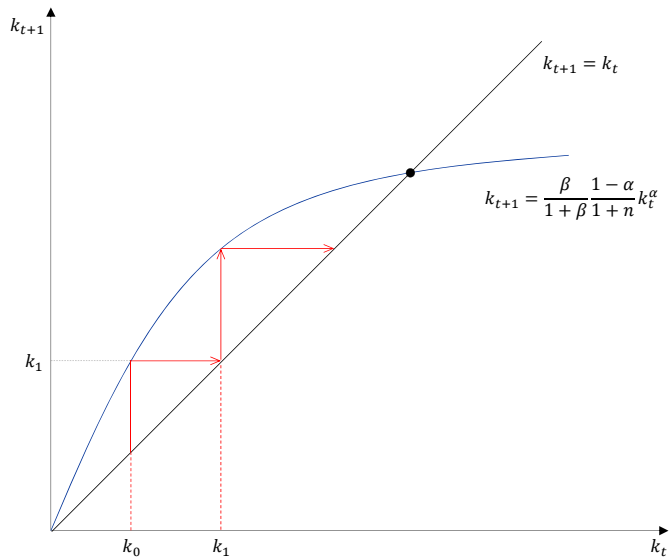
$$\alpha = 0.36, \quad \beta \approx 0.40, \quad n = 0.10.$$

	Formula	Value
A	$\frac{\beta(1-\alpha)}{(1+\beta)(1+n)}$	0.166
k^*	$A^{1/(1-\alpha)}$	0.061
r^*	$\alpha(k^*)^{\alpha-1}$	2.166
n	population growth	0.100

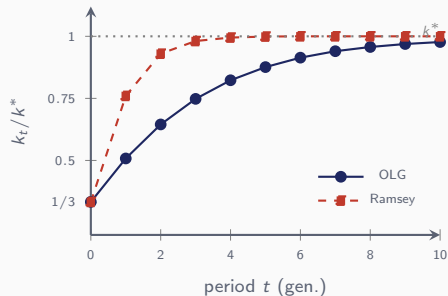
$$r^* \approx 2.17 \gg n = 0.10 \text{ (generational)}$$

$\Rightarrow r_{\text{ann}} \approx 3.9\%$, $n_{\text{ann}} \approx 0.32\%$: **dynamically efficient**. To get $r^* < n$ need very high β (patient) or very low α .

Parametric Example: Cobb-Douglas + Log Utility



Transition Dynamics: OLG vs. Ramsey



Both paths start at $k_0 = k^*/3$ and both reach k^* . The difference is *not* the speed of k_t but the **jump** in c_t at $t = 0$: Ramsey yes, OLG no.

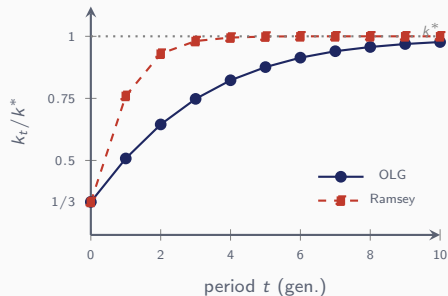
Ramsey — 2D system in (k_t, c_t) :

$$\begin{cases} (1+n)k_{t+1} = f(k_t) + (1-\delta)k_t - c_t \\ u'(c_t) = \beta[1 + f'(k_{t+1}) - \delta]u'(c_{t+1}) \end{cases}$$

+ TVC: $\lim_t \beta^t u'(c_t) k_{t+1} = 0$.

Linearised system has eigenvalues $\mu_1 \in (0, 1)$, $\mu_2 > 1$. TVC kills $\mu_2 \Rightarrow c_0$ **jumps** onto saddle path. Persistence of k : μ_1 .

Transition Dynamics: OLG vs. Ramsey



Both paths start at $k_0 = k^*/3$ and both reach k^* . The difference is *not* the speed of k_t but the **jump** in c_t at $t = 0$: Ramsey yes, OLG no.

Ramsey — 2D system in (k_t, c_t) :

$$\begin{cases} (1+n)k_{t+1} = f(k_t) + (1-\delta)k_t - c_t \\ u'(c_t) = \beta[1 + f'(k_{t+1}) - \delta]u'(c_{t+1}) \end{cases}$$

+ TVC: $\lim_t \beta^t u'(c_t) k_{t+1} = 0$.

Linearised system has eigenvalues $\mu_1 \in (0, 1)$, $\mu_2 > 1$. TVC kills $\mu_2 \Rightarrow c_0$ **jumps** onto saddle path. Persistence of k : μ_1 .

OLG — 1D system in k_t alone:

$$(1+n)k_{t+1} = s(\omega(k_t)) = \frac{\beta}{1+\beta}(1-\alpha)k_t^\alpha \equiv g(k_t)$$

No TVC needed. Linearise:

$$k_{t+1} - k^* \approx g'(k^*)(k_t - k^*), \quad g'(k^*) = \alpha.$$

Convergence rate = capital share.

The Command Optimum

Setup: Social Welfare and Resources

Social welfare function — planner weights cohorts by $(1 + R)^{-t}$:

$$\mathcal{U} = \beta u(c_0^2) + \sum_{t=0}^{T-1} \frac{1}{(1 + R)^t} [u(c_t^1) + \beta u(c_{t+1}^2)].$$

R : intergenerational discount rate. $R = 0 = \text{equal weights on all cohorts}$.

Aggregate resource constraint:

$$K_t + F(K_t, N_t) = K_{t+1} + N_t c_t^1 + N_{t-1} c_t^2.$$

Dividing by N_t and using $N_{t-1} = N_t/(1 + n)$:

$$\underbrace{k_t + f(k_t)}_{\text{resources per young}} = \underbrace{(1 + n)k_{t+1}}_{\text{cap. for next cohort}} + c_t^1 + \frac{1}{1+n} c_t^2.$$

Lagrangian and First-Order Conditions

Form the Lagrangian:

$$\mathcal{L} = \beta u(c_0^2) + \sum_{t=0}^{T-1} \frac{1}{(1+R)^t} \left\{ u(c_t^1) + \beta u(c_{t+1}^2) + \lambda_t [k_t + f(k_t) - (1+n)k_{t+1} - c_t^1 - \frac{c_t^2}{1+n}] \right\}.$$

Maximising w.r.t. c_t^1 , c_{t+1}^2 , k_{t+1} :

$$\lambda_t = u'(c_t^1), \quad (1)$$

$$\beta u'(c_{t+1}^2) = \frac{1}{1+R} \frac{\lambda_{t+1}}{1+n}, \quad (2)$$

$$\lambda_t = \frac{1}{1+R} \frac{\lambda_{t+1}}{1+n} [1 + f'(k_{t+1})]. \quad (3)$$

(1): marginal utility of young = shadow value of resources.

(2): marginal utility of old, discounted by β , equals the shadow price *next period* adjusted by planner discount and cohort size.

(3): Euler for capital — saving one unit today costs λ_t and returns $[1 + f'(k_{t+1})]$ units of resources next period.

Planner's Euler Equation and Steady State

Combining (1)–(3) gives the **planner's Euler**:

$$u'(c_t^1) = \beta[1 + f'(k_{t+1})]u'(c_{t+1}^2).$$

Identical in form to the household Euler — only the *price* of intertemporal trade differs (here $1 + f'(k_{t+1})$ is the social rate of return, not a market r).

Steady state ($c_t^1 = c^1$, $c_t^2 = c^2$, $k_t = k$):

$$\beta u'(c^2) = \frac{1}{1+R} \frac{u'(c^1)}{1+n}, \quad 1 + f'(k) = (1+R)(1+n)$$

Expanding $(1+R)(1+n) = 1 + R + n + Rn$ and neglecting Rn :

$$f'(k) \approx R + n.$$

If $R = 0$ (equal weights), $f'(k) = n$: MPK equals population growth. This recovers the *golden rule* for stationary per-capita consumption.

Pareto Efficiency and the Golden-Rule Criterion

Total per-capita consumption (young + old, weighted by population):

$$c \equiv c^1 + \frac{c^2}{1+n}.$$

In steady state the resource constraint collapses to

$$f(k) - nk = c. \quad (4)$$

Differentiating:

$$\frac{dc}{dk} = f'(k) - n.$$

- $f'(k) > n$: lowering k *reduces* stationary consumption $\Rightarrow$ economy is **dynamically efficient**.
- $f'(k) < n$: lowering k *raises* stationary consumption $\Rightarrow$ economy is **dynamically inefficient**.

Is the Decentralised Equilibrium Efficient?

Decentralised steady state: $(1+n)k = s(w(k), r(k))$.

Cobb-Douglas + log utility:

$$k_{t+1} = \frac{s_t}{1+n} = \frac{\beta}{1+\beta} \frac{1-\alpha}{1+n} k_t^\alpha \Rightarrow k = \left(\frac{\beta}{1+\beta} \frac{1-\alpha}{1+n} \right)^{1/(1-\alpha)}.$$

In this steady state,

$$r = f'(k) \gtrless n \iff \frac{\alpha}{1-\alpha} \frac{1+\beta}{\beta} \gtrless \frac{n}{1+n}$$

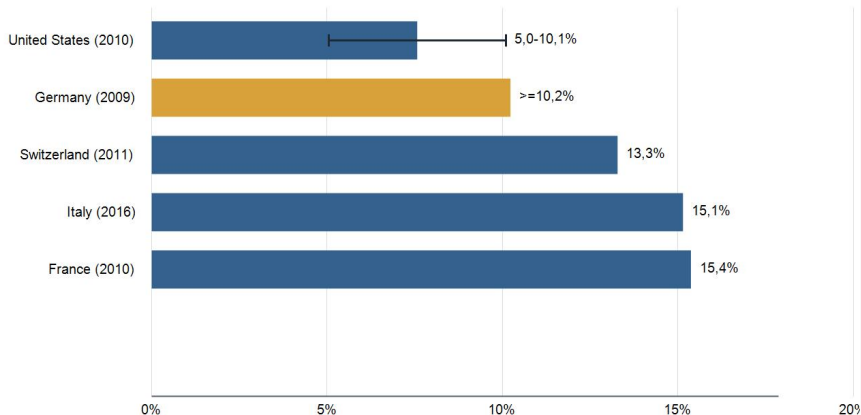
With $\alpha \approx 0.36$, $\beta \approx 0.40$ (30-yr), $n = 0.10$:

$\frac{\alpha}{1-\alpha} \frac{1+\beta}{\beta} = \frac{0.36}{0.64} \cdot \frac{1.4}{0.4} \approx 1.97$, whereas $\frac{n}{1+n} \approx 0.09$.

The LHS dwarfs the RHS $\Rightarrow$ **Pareto inefficiency is unlikely**. Getting $f'(k) < n$ would require implausibly high β or low α .

Altruism: Restoring the Command Optimum

Inheritance Flows in Selected Countries



Published literature values are mostly shares of national income; GDP shares are approximated using same-year World Bank GNI/GDP ratios.
Germany is a conservative lower bound; the United States is shown as a literature range benchmark.

Approximate inheritance flow (% of GDP)

Altruism: Restoring the Command Optimum

Parent's recursive utility (values children's lifetime utility at rate R):

$$V_t = u(c_t^1) + \beta u(c_{t+1}^2) + \frac{1}{1+R} V_{t+1}.$$

Solving it forward:

$$V_t = \sum_{i=0}^{\infty} \frac{1}{(1+R)^i} [u(c_{t+i}^1) + \beta u(c_{t+i+1}^2)].$$

This is *exactly* the planner's objective (up to the initial-old term). Altruism turns the dynasty into an infinitely-lived decision-maker.

The Altruistic Household's Problem

Budget with *non-negative* bequests $beq_t \geq 0$:

$$c_t^1 + s_t = w_t + beq_t,$$

$$c_{t+1}^2 + (1+n)beq_{t+1} = (1+r_{t+1})s_t.$$

(Negative bequests would mean leaving debt to children: ruled out.) Intertemporal form:

$$c_t^1 + \frac{c_{t+1}^2 + (1+n)beq_{t+1}}{1+r_{t+1}} = w_t + beq_t.$$

Maximising the obj. fct. s.t. intertemporal constraint:

$$u'(c_t^1) = \beta[1+r_{t+1}]u'(c_{t+1}^2),$$
$$u'(c_t^1) \frac{1+n}{1+r_{t+1}} \begin{cases} = \frac{u'(c_{t+1}^1)}{1+R} & \text{if } beq_{t+1} > 0, \\ \geq \frac{u'(c_{t+1}^1)}{1+R} & \text{if } beq_{t+1} = 0. \end{cases}$$

Equivalence with the Command Optimum

Result (Barro-style equivalence)

If bequests are **interior** ($beq_{t+1} > 0$) and factor prices clear at $r_{t+1} = f'(k_{t+1})$, the altruistic household's FOCs coincide with the planner's FOCs: $\Rightarrow$ the decentralised economy *is* Pareto-efficient.

When does the equivalence break?

- **Corner** $beq_{t+1} = 0$: the non-negativity constraint binds; parents would *like* to “take from children” but cannot.
- Empirical evidence (Altonji–Hayashi–Kotlikoff 1997): implications of altruism for intergenerational *risk sharing* are rejected in the data. The operative-bequest assumption is strong.

The planner's problem under log utility and $R = 0$ reduces (using $c_t^2 = \beta(1+n)c_t^1$ from) to a **first-order system in the state-control pair** (k_t, c_t^1):

$$\begin{aligned} k_{t+1} &= \frac{k_t + k_t^\alpha - (1 + \beta)c_t^1}{1 + n}, \\ c_{t+1}^1 &= c_t^1 \frac{1 + \alpha k_{t+1}^{\alpha-1}}{1 + n}. \end{aligned} \tag{5}$$

k_t is a *state* (given by history), c_t^1 is a *jump variable* (chosen by the planner).

Given k_0 , which c_0^1 should the planner choose? Not all values are admissible — only those consistent with the TVC $\lim_{t \rightarrow \infty} (1 + R)^{-t} \lambda_t k_{t+1} = 0$.

Saddle-Path Structure via Linearisation

Linearise the planner's system around the steady state (k^*, c^{1*}) where $k^* = (\alpha/n)^{1/(1-\alpha)}$ and $c^{1*} = [f(k^*) - nk^*]/(1 + \beta)$:

$$\begin{pmatrix} k_{t+1} - k^* \\ c_{t+1}^1 - c^{1*} \end{pmatrix} = J \begin{pmatrix} k_t - k^* \\ c_t^1 - c^{1*} \end{pmatrix}, \quad J = \begin{pmatrix} 1 & -\frac{1+\beta}{1+n} \\ \frac{c^{1*} f''(k^*)}{1+n} & 1 + \frac{c^{1*} f''(k^*)}{1+n} \left(-\frac{1+\beta}{1+n}\right) \end{pmatrix}.$$

For our calibration ($\alpha = 0.36, \beta = 0.40, n = 0.10$):

$$J \approx \begin{pmatrix} 1 & -1.273 \\ -0.0074 & 1.009 \end{pmatrix}, \quad \det J = 1, \quad \text{tr } J \approx 2.009.$$

The eigenvalues

$$\mu_s = 0.9076 < 1 < 1.1018 = \mu_u, \quad \mu_s \cdot \mu_u = 1.$$

Saddle $\Rightarrow$ unique stable manifold.

The TVC-compatible initial condition. Choose c_0^1 so that $(k_0 - k^*, c_0^1 - c^{1*})$ lies along the *stable eigenvector* v_s . Any other choice places weight on the unstable direction and violates the TVC.

Computing the Saddle Path: Eigenvector Projection

Step 1. Find v_s (eigenvector of J for $\mu_s < 1$ that is $(J - \mu_s I)v_s = 0$). Its *slope* gives the saddle-path policy rule:

$$s \equiv \frac{v_s^{(c)}}{v_s^{(k)}} \Rightarrow c^{1,SP}(k) = c^{1*} + s(k - k^*).$$

Step 2. For our calibration, $s = 0.0726$ (young consumption rises by ≈ 0.07 units for each unit rise in k along the saddle path).

Step 3. Given $k_0 = 0.7k^* = 5.18$:

$$c_0^1 = c^{1*} + 0.0726(k_0 - k^*) = 0.9397 + 0.0726 \cdot (-2.22) = 0.7785.$$

Step 4. Iterate the nonlinear system from $(k_0, c_0^1) = (5.180, 0.779)$. Path converges to (k^*, c^{1*}) at rate $\mu_s = 0.91$ per generation.

Decentralised vs Planner's OLG: The Punchline

	Decentralised OLG	Planner's OLG
Dynamic system	1D: $k_{t+1} = g(k_t)$	2D: (k_t, c_t^1)
Eigenvalues	one: $\alpha = 0.36$	two: 0.91, 1.10
Jump variable	none	c_t^1
TVC	absent	restored
Saddle path	trivial	non-trivial
Convergence persistence	0.36 per gen.	0.91 per gen.
Who optimises long-run?	no one	planner

Summary

Summary: Key Results — Lecture 2

1. **Exchange $\rightarrow$ production.** Euler equation identical. $w_t = \omega(k_t)$, $r_t = \rho(k_t)$ endogenous.
2. $s(w, r)$. $\partial s / \partial w \in (0, 1)$ always. $\partial s / \partial r \geq 0 \Leftrightarrow \text{IES} \geq 1 \Leftrightarrow$ backward-bending offer curve (L1) $\Leftrightarrow$ poverty traps (L2).
3. **Offer curve in (c^1, c^2) space.** Budget slope $-(1 + r^*)$ vs. feasibility slope $-(1 + n)$. Samuelson: $r^* < n \Leftrightarrow$ budget shallower than feasibility. Golden Rule ($r^* = n$) is the L2 analog of the L1 monetary equilibrium.
4. **Transition map $g(k_t)$.** Unique stable SS in CD+log ($g' = \alpha < 1$). Non-monotone $\Rightarrow$ poverty traps ($\text{IES} < 1$).
5. $r^* \geq n \Leftrightarrow$ **dynamic efficiency.** Production counterpart of L1's $\bar{r} < 0$ (Samuelson).
6. **PAYG = Social Ponzi (L1).** Return $1 + n$ replaces $1 + \bar{r}$. Dominates iff $r^* < n$.
7. $r < g$: impossible in Ramsey (TVC), possible in OLG (no aggregate TVC).

-  Abel, A.B., Mankiw, N.G., Summers, L.H. & Zeckhauser, R.J. (1989). Assessing dynamic efficiency. *Review of Economic Studies*, 56(1), 1–19.
-  Altonji, J.G., Hayashi, F. & Kotlikoff, L.J. (1997). Parental altruism and inter vivos transfers. *Journal of Political Economy*, 105(6), 1121–1166.
-  Barro, R.J. (1974). Are government bonds net wealth? *Journal of Political Economy*, 82(6), 1095–1117.
-  Blanchard, O. (2019). Public debt and low interest rates. *American Economic Review*, 109(4), 1197–1229.
-  de la Croix, D. & Michel, P. (2002). *A Theory of Economic Growth: Dynamics and Policy in OLG*. Cambridge University Press.

-  Diamond, P.A. (1965). National debt in a neoclassical growth model. *American Economic Review*, 55(5), 1126–1150.
-  Gale, D. (1973). Pure exchange equilibrium of dynamic economic models. *Journal of Economic Theory*, 6(1), 12–36.
-  Balasko, Y. & Shell, K. (1980). The overlapping-generations model, I. *Journal of Economic Theory*, 23(3), 281–306.